(2) Assignment operator :—

=> The purpose of Assignment operator is that "To assign or transfer Right Hand size (RHS) value/ Expression value to the left Hand side (LHS) variable".

=> In the symbol for Assignment operator is single equal to (=)

=> In python programming, we can use Assignment operator in two ways:-

(1) single Line Assignment.
(2) multi line Assignment.

(1) Single line Assignment

Syntax : LHS Varname = RHS value
LHS varname = RHS expression.

=> with single line Assignments at a time we can Assign one RHS value/ expression to the single LHS variable Name.

Examples

```
>>> a=10
>>> b=20
>>> c=a+b
>>> print(a,b,c) --- 10,2030
```

(2) multiline Assignment :-

syntax: var1, var2 . . . . var-n = val1, val2 . . . . val-n
var1, var2 . . . . varn = expr1, expr2 . . . exprn

=) Here, the values of val1, val2 . . . val-n are assigned to var1, var2 . . . . var-n respectively.

=) Here, the values of expr1, expr2 . . . exprn are assigned to var1, var2 . . . . var-n respectively.

Examples:

```
>>> a, b = 10, 20
>>> print (a, b) — — —  10 20
>>> c, d, e = a + b, a - b, a * b
>>> print (c, d, e) — — — 30  -10  200
>>> sno, sname, marks = 10, "Rossum", 34.56
>>> print (sno, sname, marks) — — — 10 Rossum 34.56 .
```

```
>>> a, b = 10, 20
>>> print (a, b) — — —  10 20
>>> a, b = b, a    # swapping logic
>>> print (a, b) — — —  20 10 .
```

programming for swapping of two Numbers:-

```
a, b = input ("Enter value of a :"), input ("Enter value of b :")
        # multiline Assignment
print ("*" * 50)
print ("original value of a = {}". format (a))
print ("original value of b = {}". format (b))
print ("*" * 50)
# swapping or interchanging.
a, b = b, a
print ("swapped value of a = {}". format (a)).
print ("swapped value of b = {}". format (b)).
print ("*" * 50).
```

```
# program for swapping of two values.
a,b=input ("enter value of a:"), input ("Enter value of b:")
print ("*"*50).
print ("original value of a = {}".format(a))
print ("original value of b = {}".format (b))
print ("*"*50).
#swapping or interchanging logic.
t=a
a=b
b=t
print ("swapped value of a = {}".format(a))
print ("swapped value of b = {}".format (b))
print ("*"*50)
```

```
#program for swapping two integer values only.
a,b=(input ("Enter value of a:")),(input (" Enter value of b:")),
print ("*"*50)
print ("original value of a = {}".format (a)).
print ("original value of b = {}".format (b))
print ("*"*50)
#swapping or interchanging.
a=a+b
b=a-b
a=a-b
print ("swapped value of a = {}".format (a))
print ("swapped value of b = {}".format (b))
print ("*"*50)
```

Day (29)

(3) Relational operators (Comparision operator)

→ The purpose of Relational operators is that "To compare Two values".

→ if two values connected Relational operators then it is called "Relational Expression".

→ The Relational Expression is either 'True' or 'False'

→ The Relational Expression is also called Test condition and whose result can be either True or false.

→ In python programming, we have 6 types of Relational operators. They are given in the following Table.

SLNo.	symbol	Meaning	Example
1.	>	Greater than	print(10>5) --- True Here 10>5 is called Relational expression or Test cond.
2.	<	less than	print(10<20) --- True print(200<80) --- false.
3.	==	equality (double equal to)	print(10==20) --- false print(100==100) --- True.
4.	!=	Not equal to	print(10!=20) --- True print(100!=100) --- false.
5.	>=	Greater than or equal to	print(10>=2) --- True print(10>=20) --- false
6.	<=	less than or equal to	print(10<=20) --- True print(10<=2) -- false.

```
# program for Demonstrating Relational operators.
    a,b = int(input("Enter value of a":)), int(input("Enter
                                    a value of b:")).
print("*" *50)
print("Results of Relational operators")
print("*"*50)
print("\t\t {} > {} = {}". format(a,b,a>b))
print("\t\t {} < {} = {}". format(a,b,a<b))
print("\t\t {} == {} = {}". format(a,b,a==b))
print("\t\t {} != {} = {}". format (a,b, a!=b))
print("\t\t {} >= {} = {}". format (a,b, a>=b))
print("\t\t {} <= {} = {}". format (a,b, a<=b))

print("*"*50)
```

2.Assignment Operators

in pyhton programming we can use asiignment operators in two ways:

- 1.single line Assignment:-
- 2.multi line Assignment:-

1. single line Assignment operators:

- syntax:- LHS varname=RHS value (or) LHS varname=RHS expression

In [2]:
```
a=10
b=20
c=a+b
print(a,b,c)
```

```
10 20 30
```

2.multiline Assignment operators:

- syntax: var1,var2....varn=val1,val2........valn
- or
- `var1,var2.....varn=expr1,expr2......exprn`

In [3]:
```
a,b=10,20
```

```
print(a,b)
```

```
10 20
```

In [5]:
```
c,d,e=a+b,a-b,a*b
print(c,">",d,">",e)
```

```
30 > -10 > 200
```

In [6]:
```
sno,sname,marks=10,"rossum",34.56
print(a,b)
a, b =b,a
print(a,b)
```

```
10 20
20 10
```

programming for swapping of two numbers:-

In [7]:
```
a,b=input("enter value of a:"),input("enter value of b:")
print("="*50)
print("original value of a={}".format(a))
print("original value of b={}".format(b))
print("="*50)
#swapping or interchanging
a,b=b,a
print("swapped value of a={}".format(a))
print("swapped value of b={}".format(b))
print("="*50)
```

```
==================================================
original value of a=54
original value of b=78
==================================================
swapped value of a=78
swapped value of b=54
==================================================
```

programming for swapping of two values:

In [9]:
```
a,b=input("enter value of a:"),input("enter value of b:")
print("="*50)
print("value of a={}".format(a))
print("value of b={}".format(b))
print("="*50)
#swapping or interchanging logic
t=a
a=b
b=t
print("swapped value of a={}".format(a))
print("swapped value of b={}".format(b))
print("="*50)
```

```
==================================================
value of a=50
value of b=60
==================================================
swapped value of a=60
swapped value of b=50
==================================================
```

program for swapping two integer values only:-

In [16]:
```
a,b=int(input("enter value of a:")),int(input("enter value of b:"))
print("="*50)
print("original value of a={}".format(a))
print("original value of b={}".format(b))
print("="*50)
#swapping or interchanging
a=a+b
b=a-b
a=a-b
print("swapped value of a={}".format(a))
print("swapped value of b={}".format(b))
print("="*50)
```

```
==================================================
original value of a=12
original value of b=8
==================================================
swapped value of a=8
swapped value of b=12
==================================================
```

3. Relational operators(comparison operator)

program for demonstrating relational operators.

In [18]:
```
a,b=int(input("enter value of a:")),int(input("enter value of b:"))
print("="*50)
print("Results of Relational operators")
print("="*50)
print("\t \t{}>{}={}".format(a,b,a>b))
print("\t \t{}<{}={}".format(a,b,a<b))
print("\t \t{}=={}={}".format(a,b,a==b))
print("\t \t{}!={}={}".format(a,b,a!=b))
print("\t \t{}>={}={}".format(a,b,a>=b))
print("\t \t{}<={}={}".format(a,b,a<=b))
```

```
==================================================
Results of Relational operators
==================================================
                1012>1014=False
                1012<1014=True
                1012==1014=False
                1012!=1014=True
                1012>=1014=False
                1012<=1014=True
```

In [ ]: